Singapore University of Technology and Design

50.039 Theory and Practices of Deep Learning

Final Report

Real-time Aircraft Engine Abnormal Remaining Useful Life (RUL) Prediction

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Group 5

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1. Introduction

The aviation industry faces significant challenges regarding safety and operational efficiency. Predictive maintenance (PdM) is essential to identify “abnormal” engine behaviour before catastrophic failure occurs, reducing costs from unscheduled repairs and fatal accidents.

While traditional maintenance relied on scheduled checks. PdM utilizes historical and real-time sensory data to model degradation patterns. By analysing temporal dependencies, models can predict the state of a component based on its operational history.

A key challenge in this domain is the inherently imbalanced nature of engine health data (i.e. most operational cycles represent normal, healthy engine states, while only a small fraction captures the critical degradation window preceding failure). This project therefore frames the problem as anomaly detection: using regression-based RUL prediction as an intermediate representation to ultimately classify whether an engine is entering a dangerous operational state, where anomalous (near-failure) samples are significantly outnumbered by normal ones.

2. State-of-The-Art

A key direction in the literature is the convolution-Transformer family. For example, the GCU-Transformer integrates a gated convolutional unit before the Transformer encoder so that local temporal patterns are extracted before self-attention models global context (Y. Mo, Q. Wu, X. Li, and B. Huang, 2021). This is useful especially in sensor data because degradation often appears first as short-term changes in sensor trends rather than as a clear global pattern. In the same line, DSFormer introduces dual-scale temporal modeling and convolutional components to capture both short-term and longer-horizon information, which helps when the degradation trajectory changes across multiple time scales (J. Li et al., 2024). Another strong variant is AGCTE, which combines an adaptive graph convolutional network with a Transformer encoder to better model inter-sensor relationships (J. Li et al., 2024). More broadly, hybrid designs that place convolution before Transformer stages are motivated by the idea that convolution is better at extracting local patterns in early stages (Tan et al., 2021).

3. Implementation

3.1. Data Exploration (EDA)

3.1.1. Dataset Overview

The analysis uses the NASA CMAPSS (Commercial Modular Aero-Propulsion System Simulation) dataset, specifically one main sub-dataset: FD001. FD001 contains time-series sensor readings from turbofan engines, structured with 26 columns: an engine ID (id), an operational cycle counter (cycle), three operational setting variables (setting1, setting2, setting3), and 21 sensor readings (s1 through s21). Additionally, accompanied with the data would be the Remaining Useful Life target feature that is constructed from taking the maximum cycles minus current cycles for the training sets, whereas for it is provided directly as a file for each engine in the test set.

For this project, we will be focusing on FD001 as our main dataset to train and finetune our models.

Dataset	**Train Rows**	**Test Rows**	**Mean Cycle**	**Max Cycle**
FD001	20,631	13,096	108.8	362

Figure 3.1.1.1: Size of Datasets

Subset	Operation Conditions	Fault modes	Use case
FD001	1	1	Baseline single condition case

Figure 3.1.1.2: Characteristics of FD001

	FD001		
	min	mean	max
id	1.0000	51.5066	100.0000
cycle	1.0000	108.8079	362.0000
setting1	-0.0087	-0.0000	0.0087
setting2	-0.0006	0.0000	0.0006
setting3	100.0000	100.0000	100.0000
s1	518.6700	518.6700	518.6700
s2	641.2100	642.6809	644.5300
s3	1571.0400	1590.5231	1616.9100
s4	1382.2500	1408.9338	1441.4900
s5	14.6200	14.6200	14.6200
s6	21.6000	21.6098	21.6100
s7	549.8500	553.3677	556.0600
s8	2387.9000	2388.0967	2388.5600
s9	9021.7300	9065.2429	9244.5900
s10	1.3000	1.3000	1.3000
s11	46.8500	47.5412	48.5300
s12	518.6900	521.4135	523.3800
s13	2387.8800	2388.0962	2388.5600
s14	8099.9400	8143.7527	8293.7200
s15	8.3249	8.4421	8.5848
s16	0.0300	0.0300	0.0300
s17	388.0000	393.2107	400.0000
s18	2388.0000	2388.0000	2388.0000
s19	100.0000	100.0000	100.0000
s20	38.1400	38.8163	39.4300
s21	22.8942	23.2897	23.6184

Figure 3.1.1.3: Characteristics of Features in FD001

3.1.2. Sensor Visualisation

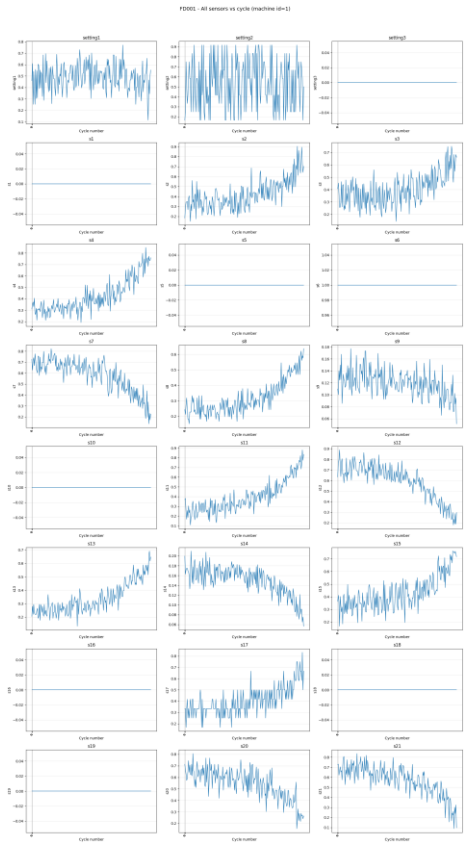


Fig 3.1.2: Sensor Readings Across Cycles for ID 1 Engine for FD001

To understand degradation patterns, all 21 sensor signals were plotted against operational cycles for individual engines. This per-engine visualisation revealed that different sensors exhibit varying degrees of monotonic trends as engines approach failure — some increasing (e.g., s12, s7), some decreasing (e.g., s11, s4), while others remain flat throughout the lifecycle. More importantly, this visual inspection tells us that for our main dataset FD001, there may exist some features that do not contribute significantly to engine degradation, which we may remove to remove noise in the dataset that do not contribute meaningfully. This is further studied in the Section 3.1.4 for a quantitative view.

3.1.3. Cycle Distribution

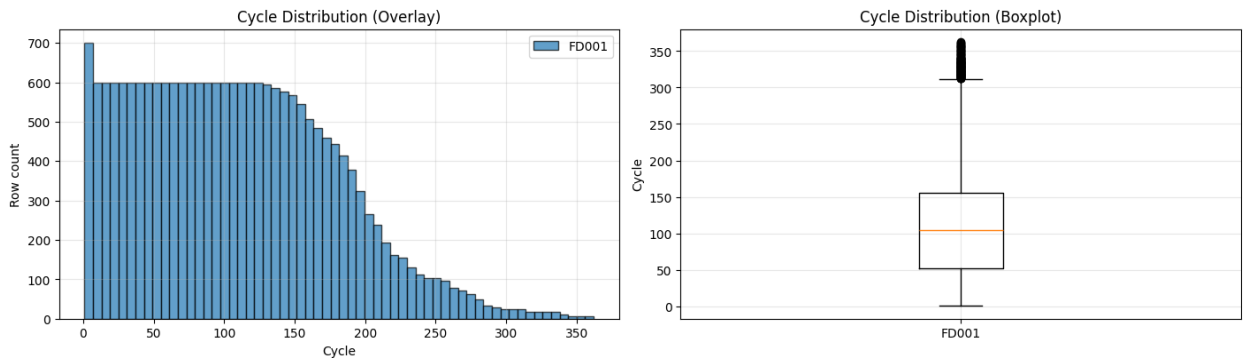


Fig 3.1.3.1: Cycle Distribution for FD001 Dataset

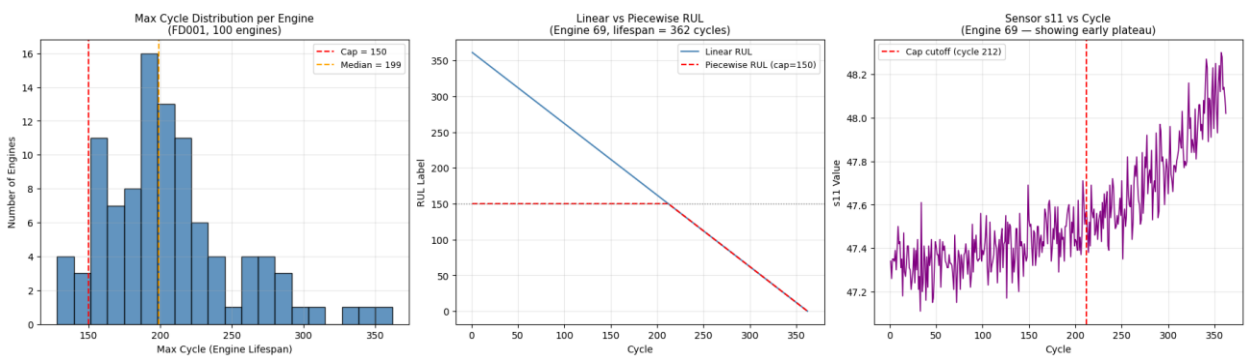


Fig 3.1.3.2 Max Cycle Distribution and Engine 69 Example in FD001

From the cycle distribution, we can determine the distribution of maximum engine cycles for all engines for both datasets. FD001 shows a right-skewed distributions, which indicates that a large proportion of engines fail relatively early as seen from the drop off at around ~125 cycles. This finding suggests to us that almost every engine has a long pre-degradation phase where most of the sensor readings are unlikely to contribute to any change of RUL.

For a clearer example, we can look at Engine ID 69 that has a lifespan of 362 cycles. We can see that the corresponding sensor reading is flat from cycle 1 to cycle 212 before beginning its characteristic decline. Under linear RUL, rows in that flat zone are labelled RUL = 362 down to

150, which represents a span of 212 unique labels attached to sensor readings that may be almost the same, which forces model to learn arbitrary distinctions from a noisy plateau.

This cycle distribution shows us a specific characteristic of the engine behaviour within FD001 dataset which is that most engines spend a significant proportion of life in a state that is no different than new/not degrading. This distribution also reveals the class imbalance inherent to the dataset: if failure is defined as RUL ≤ 30 cycles, only ~15% of training rows fall into the anomalous class, with the remaining ~85% representing normal operation.

3.1.4. Variance Analysis

	FD001_train_variance
setting1	0.0158024
setting2	0.0596426
setting3	0
s1	0
s2	0.0226859
s3	0.017866
s4	0.0230841
s5	0
s6	0.0192928
s7	0.0203139
s8	0.0115678
s9	0.00981855
s10	0
s11	0.0252748
s12	0.024731
s13	0.0111858
s14	0.00969091
s15	0.0208241
s16	0
s17	0.0166574
s18	0
s19	0
s20	0.0196318
s21	0.0223432

Fig 3.1.4: Variance Analysis for FD001

In addition to looking at the patterns of data, we also looked at the variance of each setting and feature. Based on the prior visual analysis of Engine ID 1, we notice that some of the features have no signal at all, which we presume is what creates the singular operational condition of FD001 dataset. This variance analysis hence informs us exactly which variables contribute towards changes in RUL, motivating our selection of features in Section 3.3.1.

3.1.5. Correlation Analysis

First, we conducted a correlation analysis of each variable with respect to RUL to determine which features were potentially highly influential in resulting in variations in RUL.

FD001 - Feature correlation with RUL (sorted by |corr|):

	corr_with_RUL	abs_corr_with_RUL
s11	-0.775230	0.775230
s4	-0.757157	0.757157
s12	0.748870	0.748870
s7	0.733021	0.733021
s15	-0.720858	0.720858
s21	0.707334	0.707334
s20	0.704626	0.704626
s17	-0.680829	0.680829
s2	-0.678458	0.678458
s3	-0.655030	0.655030
s8	-0.624568	0.624568
s13	-0.624034	0.624034
s9	-0.462151	0.462151
s14	-0.369753	0.369753
s6	-0.108289	0.108289
setting2	-0.007091	0.007091
setting1	-0.005556	0.005556

Fig 3.1.5: Correlation of features with RUL for FD001

From the above analysis, we can observe that there are many variables in FD001 have a high correlation to RUL.

Next, we wanted to find which features had a high correlation with one another to identify potential features that may result in overfitting of our dataset. This would help us in feature selection down the line to ensure as much independence between features as possible.

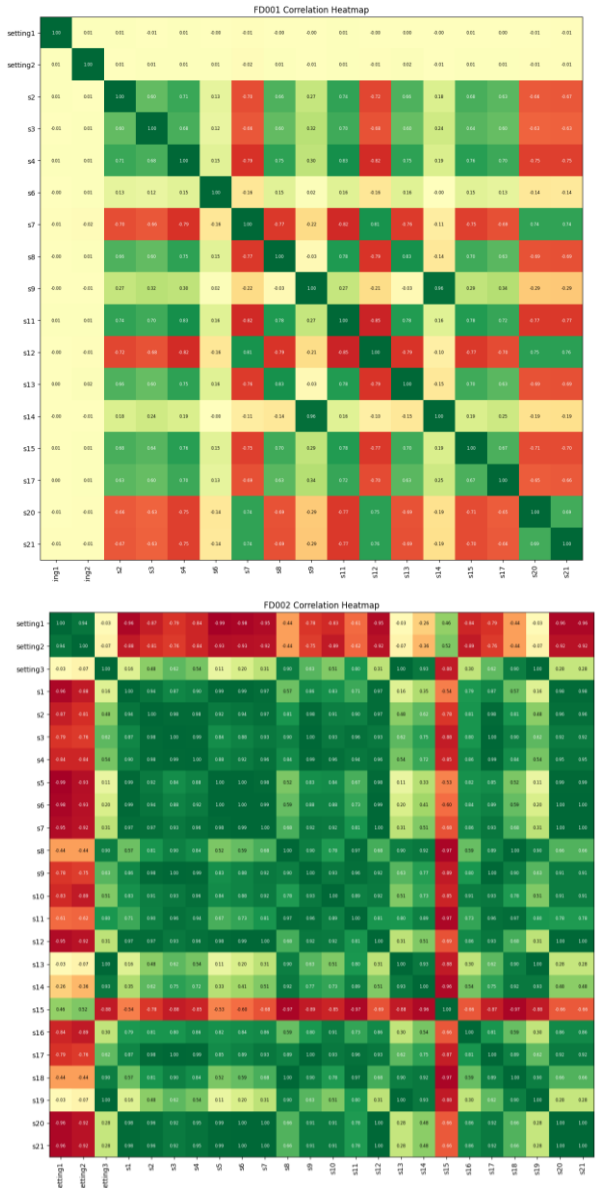


Fig 3.1.5: Interfeature Correlation Analysis for FD001

From the correlation heatmap above where green represents highly positive correlations and red represents highly negative correlations, we can identify pairs of features in FD001 such as s14 and s9 as well as s12 and s11 (>0.85 correlation). We can use this knowledge on top of the RUL correlation analysis to help determine which features to remove in Section 3.3.2.

3.2. Data Preprocessing

3.2.1. RUL Label Reconstruction

Rather than using a raw linear countdown from maximum cycle to zero, RUL can be reconstructed as RUL = min(max_cycle – current_cycle, 125). This collpases all early-life labels that would exceed 125 down to a flat ceiling of 125. As discussed in Section 3.1.3, engines do not degreade the moment they are switched on. Degradation signals emerge meaningfully only in the last cycles of an engine's life. This would mean that for example an engine at cycle 5 with lifespan of 300 cycles (RUL = 295) would have near identical sensor readings to another equivalent engine at cycle 50 (RUL = 250). Models trained on this will be forced to learn a distinction between RULs that does not exist within input features. A piecewise cap eliminates this by treating any engine beyond 125 cycles as just a state of "healthy" to ensure uniform treatment across all engines in the earlier phases.

The above effect of capping at 125 ensures that engines at healthy plateaus receive same label. This overall helps the models learn faster in degradation regions which contains most of the feature variations and reducing training capacity wasted on high RUL values with similar sensor readings.

An additional motivation to choosing RUL cap at 125 would be that it is the standard adopted by the vast majority of published CMAPSS benchmarks, which allows our RMSE to be computed on the same label space as other papers (Heimes, 2008). This ensures that the model results are more comparable against SOTA equivalents.

3.2.2. Min-max Normalisation of Features

For each numerical feature column (all settings and sensor readings), we apply a min-max normalisation, which maps every feature into the range [0,1]. Critically, the min and max are computed only from the training set and applied to both the training and test sets since test set statistics are unseen and normalisation can only be applied using seen data.

The purpose of min-max normalisation is to remove scale dominance since every feature has their own scale. As seen in Figure 3.1.1.3, an example of difference in scale would be features like s9 have a magnitude in the thousands whereas features like s10 has a small spread around 1.

Without normalisation, features with larger absolute values would numberically dominate the gradient-based learning. This means that the neural network weight updates being proportional to the input signal, sensor with values with higher scale would receive disproportionately large gradient updates compared to a sensor with a smaller scale.

Additionally, by normalising the features, each feature is now more comparable in magnitude as differences in normalised features are more directly comparable. Gradient descent would be able to converge faster as normalisation reshapes loss landscape towards a smoother geometry as a result.

3.3. Feature Engineering

3.3.1. Low-Variance Feature Removal

As discussed in Section 3.1.4, there exists a significant number of features in FD001 that have zero variance. For FD001 specifically, there are 7 features that are dropped (s1, s5, s10, s16, s18, s19, setting3) as they are constant by design of the operational condition set in FD001.

Low variance feature removal helps to elimiminate constant features that have zero contribution towards variations in RUL. Hence, this means that no matter what coefficient is assigned to these features, the feature would then contribute equally to any prediction, rendering it unworthy to keep. The key benefits of this for our project purposes would be to ensure efficiency in computation by removing the dimensionality of the input vectors, which would improve training speeds significantly.

3.3.2. Feature Pruning Using RUL and Interfeature Correlation

From Section 3.1.5, we have determined that variables s14 and s9 and s12 and s11 show high correlation with one another. To decide which one to prune exactly, we also refer to the RUL correlation analysis. Hence, we decide to prune s14 instead of s9 because s9 (-0.462151) shows a higher correlation to RUL compared to s14 (-0.369753). Between s12 and s11, both have relatively high correlations with RUL (~0.75 each) which indicates to us that we should experiment between dropping s12 and s11 rather than immediately drop one or the other.

3.4. Data Splitting, Model Saving, and Loading

To ensure robust model training and evaluation, a systematic approach was employed for data splitting, model checkpointing, and loading. This section details these critical processes.

The dataset is split into training, validation, and test sets to facilitate model development and unbiased performance evaluation. A key aspect of our strategy is to ensure that engines are not mixed between training and validation sets, preventing data leakage and providing a more realistic assessment of the model's generalization capabilities.

1. **Train-Validation Split**: The full training dataset is first split into training and validation sets. This split is performed at the **engine ID level** to ensure that all cycles for a given engine are either entirely in the training set or entirely in the validation set. Specifically, 80% of the unique engine IDs are allocated to the training set, and the remaining 20% are allocated to the validation set. This is achieved by:

    a. Identifying all unique engine IDs.
    b. Shuffling these unique IDs using a fixed random seed (np.random.default_rng(seed=42)) for reproducibility.
    c. Dividing the shuffled IDs into 80% for training and 20% for validation.
    d. Filtering the original training set to create train and val based on these allocated engine IDs.

2. **Test Set**: The dedicated test dataset is used exclusively for final model evaluation and is kept separate throughout the training and validation phases.

3. **Sequence Creation**: For each of the dataset dataframes, time-series sequences are created using a sliding window approach. This function transforms the engine's operational data into sequences of a predefined length (seq_len) and corresponding RUL labels.

4. **DataLoader Preparation**: The generated sequences (X_train, y_train, etc.) are then converted into TensorDataset objects and wrapped in DataLoaders. The train_loader is shuffled, while val_loader and test_loader are not, to maintain sequence order for evaluation. The seed_worker function is used during DataLoader initialization to ensure reproducibility of data loading.

During the training process, the model's performance on the validation set is continuously monitored. To prevent overfitting and ensure that the best performing model is retained, a checkpointing mechanism is implemented:

- The **Root Mean Squared Error (RMSE)** on the validation set is calculated at the end of each epoch.
- If the current val_rmse is lower than the best_val_rmse recorded so far, the current model's state dictionary (model.state_dict()) is saved to a specified model_save_path using torch.save().
- This ensures that only the model weights corresponding to the best validation performance are preserved.

After the training loop is complete, the best-performing model (saved during training) is loaded back into the model architecture for final evaluation on the test set. This ensures that the evaluation is performed on the model that demonstrates the best generalization capability on unseen validation data.

3.5. Model Architectures

3.5.1. Recurrent Neural Networks (RNN)

3.5.1.1. RNN

The first model employed is a two-layer RNN designed for RUL regression. The overview of our RNN architecture is as follows:

RNN Layer 1: Takes an input of size d_{in} (number of sensor features) and maps it to a hidden representation of size $h_1 = 100$, using a $\tanh$ nonlinearity with batch-first input ordering.

- **Dropout Layer 1:** Applies stochastic regularisation with dropout rate $p = 0.2$ to the full output sequence of Layer 1
- **RNN Layer 2:** Receives the regularised sequence of size $h_1 = 100$ and produces a hidden representation of size $h_2 = 50$, again using $\tanh$ nonlinearity
- **Dropout Layer 2:** Applies dropout at $p = 0.2$ to the output sequence of Layer 2
- **Temporal Pooling:** Extracts only the final time step's hidden state, which contains the encoding the full degradation history up to the last observed cycle
- **Fully Connected Output Layer:** A linear projection of the hidden state to a scalar prediction of RUL.
- **Loss Function:** Mean Squared Error (predicted RUL – target RUL) squared

We use a vanilla RNN as our natural baseline for time-series data. Engine sensor readings are inherently ordered which makes RNN suitable due to its ability to capture temporal patterns (from beginning to end) through its hidden state.

We use a two-layer stacked design to allow the network to learn abstractions. First layer focuses on capturing low-level sensor relationships and the second layer composes these relationships to higher level degradation trends. The tanh activation used keeps hidden state values bounded to ensure numerical stability which is important for regression tasks as it ensures the hidden state

fed into the linear layer would make the linear layer's job more predictable. We also use dropout between layers to mitigate overfitting.

3.5.1.2. GRU

The second model is a two-layer Gated Recurrent Unit, sharing the same overall structure as the baseline RNN but just replacing the vanilla recurrent cells with GRU cells. Otherwise, the architecture is structurally identical for consistency purposes.

- **GRU Layer 1:** Takes input of size d_{in} and maps to hidden size $h_1 = 100$, with batch-first ordering
- **Dropout Layer 1:** Stochastic regularization at $p = 0.2$ applied across the full output sequence
- **GRU Layer 2:** Receives the regularized sequence of size $h_1 = 100$ and produces hidden representations of size $h_2 = 50$
- **Dropout Layer 2:** Dropout at $p = 0.2$ applied to the output sequence of Layer 2
- **Temporal Pooling:** Final time step hidden state is extracted
- **Fully Connected Output Layer:** Linear projection of hidden state to scalar prediction of RUL
- **Loss Function:** MSE

Engine degradation in the FD001 dataset unfolds over hundreds of operational cycles, which means that models must retain relevant information across many cycles. Vanilla RNNs struggle here as their gradients diminish exponentially as they backpropagate through many time steps, which cause earlier cycle sensor patterns to become negligible on the final prediction. In contrast, GRU's gate mitigates this by selectively preserving important hidden state information across many cycles such as memory of an early anomalous sensor reading that may be a key predictor of failure. Compared to more complex LSTM, GRU achieves similar memory capacity

with fewer parameters (no cell state, output gate), which makes it computationally efficient and choice for a smaller FD001 dataset.

3.5.1.3. LSTM

The third model is a two-layer LSTM network. It follows the same structural template as the RNN and GRU models but simply uses LSTM cells instead.

- **LSTM Layer 1:** Takes input of size d_{in} and maps to hidden size $h_1 = 100$, with batch-first ordering
- **Dropout Layer 1:** Stochastic regularization at $p = 0.2$ applied across the full output sequence
- **LSTM Layer 2:** Receives the regularized sequence of size $h_1 = 100$ and produces hidden representations of size $h_2 = 50$
- **Dropout Layer 2:** Dropout at $p = 0.2$ applied to the output sequence of Layer 2
- **Temporal Pooling:** Final time step hidden state is extracted
- **Fully Connected Output Layer:** Linear projection of hidden state to scalar predictions of RUL
- **Loss Function:** MSE

LSTM introduces a dedicated cell state which is a separate memory highway alongside the hidden state. It is governed by the three gates (forget, input and output gate). This decoupling of the cell and hidden state gives LSTM finer-grained control as the cell state can carry long-range information largely intact while hidden state only exposes what is relevant at each cycle.

The LSTM's explicit cell state and three-gate mechanism make it the most capable of the RNN architectures discussed so far in terms of capturing long-range temporal dependencies. The trade-

off relative to the GRU is a higher parameter count, which results in the LSTM being the most data-hungry of the three baselines. Its comparison against the GRU on FD001, a relatively smaller size CMAPSS dataset, would directly help us assess whether this additional cell state complexity would actually yield meaningful gains.

3.5.2. Transformer

To capture the complex temporal degradation patterns of aircraft engines, we explored the use of Transformer-based models (defined as RULTransformer), a sequence-to-one regression model implemented in PyTorch. This model is designed as a sequence-to-one architecture that leverages self-attention mechanisms to capture long-range dependencies within sensor data. The overview of our RULTransformer architecture is as follows:

1. **Input Linear Layer:** Takes an input of size d_in (number of sensor features) and projects it to a higher-dimensional latent representation of size d_model = 64.
2. **Learnable Positional Encoding:** Injects sequence order information by adding a learnable parameter of shape [1, seq_len, d_model], initialized with a small normal distribution (μ=0, σ=0.02) provides the multi-head self-attention mechanism with immediate, distinct signals to differentiate individual time steps from the first epoch, significantly accelerating the model's early convergence rate.
3. **Transformer Encoder Stack:** Consists of N stacked layers (where N is a tunable parameter, default N=2), each utilizing 4 attention heads (n_head=4) and a dropout rate of p=0.1 with batch-first input ordering.

4. **Temporal Pooling:** Extracts only the final time step's hidden state from the encoder output, which acts as a summary vector of the entire engine degradation history up to the current cycle.

5. **Fully Connected Regressor Head:** A multi-layer perceptron that first reduces the hidden state from 64 to 32 neurons with a ReLU nonlinearity and p=0.1 dropout, followed by a final linear projection to a scalar prediction of RUL.

6. **Loss Function:** MSE

Unlike RNNs, which must propagate information sequentially through a chain of hidden states resulting in an O(T) path length between distant time steps, the self-attention mechanism connects any two positions in the sequence directly in O(1) path length, making it significantly more robust at capturing long-range dependencies without the risk of vanishing gradients.

We use a learnable positional encoding instead of fixed sinusoids to allow the model to adaptively learn temporal representations suited to the specific degradation patterns present in the dataset, rather than imposing a generic fixed prior that may not reflect the structure of engine health trajectories.

The Transformer Encoder stack enables the network to learn hierarchical abstractions, where lower layers capture local sensor interactions and higher layers synthesise these into global health indicators. The inclusion of ReLU and dropout in the regressor head ensures the model can capture non-linear relationships while maintaining robustness against overfitting.

3.5.3. Hybrid

Three hybrid models namely LSTM-Transformer, GRU-Transformer, and CNN-Transformer are used to address a time-series forecasting problem which requires both short-term feature extraction and long-range context achieved by using the attention mechanism. Each model combines a front-end sequence encoder with a Transformer encoder and a final regression head, allowing the network to capture degradation patterns from sensor measurements more effectively than a single architecture alone.

3.5.3.1. LSTM-Transformer

The first hybrid model combines a two-layer *Long Short-Term Memory* network with a *Transformer encoder*. The two components are compliment each other to achieve the best result. The LSTM stage is responsible for learning sequential degradation dynamics from multivariate engine sensor inputs, while the Transformer stage refines these temporal representations by modeling interactions across the full sequence. This design is suitable for RUL prediction because engine degradation often depends on both local temporal trends and longer-range dependencies.

- **LSTM Layer 1**: Takes input features of size num_features and produces hidden representations of size lstm_hidden = 64, using batch-first ordering.
- **Dropout**: Applied between stacked LSTM layers with rate 0.1 when more than one layer is used.
- **LSTM Layer 2**: Receives the hidden sequence from the previous layer and outputs a sequence of the same hidden dimension.
- **Linear Projection**: Maps the LSTM hidden dimension from 64 to Transformer embedding size d_model = 64.
- **Transformer Encoder**: Consists of 2 encoder layers, each with 4 attention heads and feedforward dimension 256.
- **Temporal Pooling**: The hidden state from the final time step is selected as the sequence summary.
- **Fully Connected Head**: A two-layer regression head maps the Transformer output to a single RUL value.
- **Loss Function**: Mean Squared Error (MSE).

3.5.3.2. GRU-Transformer

The second hybrid model shares the same overall structure as the LSTM-Transformer to ensure architectural consistency, but uses GRU cells instead of LSTM cells. The GRU is computationally lighter than the LSTM while still capturing temporal dependencies.

- **GRU Layer 1**: Takes input of size num_features and maps it to hidden size gru_hidden = 64, with batch-first ordering.
- **Dropout**: Applied between stacked GRU layers with rate 0.1 when more than one layer is used.
- **GRU Layer 2**: Processes the sequence output of the first GRU layer and produces hidden representations of size 64.
- **Linear Projection**: Projects GRU outputs to d_model = 64 for Transformer compatibility.
- **Transformer Encoder**: Uses 2 encoder layers with 4 attention heads and feedforward dimension 256.
- **Temporal Pooling**: The last time-step representation is extracted as the sequence summary.
- **Fully Connected Head**: A regression block with one hidden layer of size 128, ReLU activation, dropout, and a final linear layer.
- **Loss Function**: Mean Squared Error (MSE).

3.5.3.3. CNN-Transformer

The last hybrid model uses one-dimensional convolutional feature extractor. Instead of processing the sequence one by one, the CNN can learn the local temporal patterns such as short-term fluctuations, abrupt changes, and sensor-level abnormally. The extracyed features are then passed into the Transformer encoder to capture context across the sequence.

- **1D Convolution Layer:** Takes num_features input channels and produces num_filters = 32 feature maps using kernel size 3 with same-padding.
- **ReLU Activation:** Introduces nonlinearity after convolution.
- **Batch Normalization:** Stabilizes feature distribution and improves training robustness.
- **Dropout:** Applied with rate 0.1 to reduce overfitting.
- **Linear Projection:** Maps the convolutional feature dimension from 32 to Transformer embedding size 64.
- **Transformer Encoder:** Contains 2 encoder layers with 4 attention heads and feedforward dimension 256.
- **Temporal Pooling:** The final time-step Transformer output is used as the sequence representation.
- **Fully Connected Head:** A two-layer regression head outputs a scalar RUL estimate.
- **Loss Function:** Mean Squared Error (MSE).

3.6. Reproducibility

```python
def seed_everything(seed=1234):
    os.environ["PYTHONHASHSEED"] = str(seed)
    os.environ["CUBLAS_WORKSPACE_CONFIG"] = ":16:8"

    random.seed(seed)
    np.random.seed(seed)

    torch.manual_seed(seed)
    torch.cuda.manual_seed(seed)
    torch.cuda.manual_seed_all(seed)

    torch.backends.cudnn.deterministic = True
    torch.backends.cudnn.benchmark = False

    torch.use_deterministic_algorithms(True)
```

Fig 3.5: Seeding Function for Reproducibility

- torch.manual_seed ensures identical CPU-side weight initialization and dropout masks across runs
- np.random.seed fixes any NumPy-based data preprocessing or shuffling
- random.seed controls Python-level randomness such as dataset splits
- torch.cuda.manual_seed_all ensures GPU-side operations produce the same results across runs

As each library maintains its own internal Pseudo-Random Number Generator (PRNG) state, seeding all four together guarantees that the model starts from the same initial weights, sees data in the same order, and applies the same dropout patterns on every run, making results directly comparable and independently reproducible.

To run the provided code and reproduce the results, the following Python packages are the core requirements:

Package	Purpose
pandas	Data manipulation and analysis
numpy	Numerical operations
torch	Deep learning framework (PyTorch)
torch.nn	Neural network modules
torch.utils.data	Data loading utilities (DataLoader, TensorDataset)
sklearn.preprocessing	Data scaling (MinMaxScaler)
sklearn.metrics	Performance metrics (mean_squared_error)
math	Mathematical functions
os	Operating system interaction
random	Python's built-in random module

The full list of requirements can be found in the requirements.txt file in our GitHub repository.

4. Malfunction

While training the models, we realised that the value set for the Seed may change the results (RMSE) dramatically depending on the model. For example, when training a LSTM model, the Test RMSE can vary from 23.43 (Seed=42) to 57.17 (Seed=1234), causing 'malfunction'.

	Sequence Length	50					
	Models	GRU			LSTM		
Dataset	Seed	Train RMSE	Val RMSE	Test RMSE	Train RMSE	Val RMSE	Test RMSE
FD001_linear_rul_no_norm_0	42	57.1433	57.785	42.7017	57.1431	57.785	42.712
FD001_linear_rul_no_norm_0	1234	57.1759	56.6008	42.2878	57.1674	56.6005	42.353
FD001_linear_rul_no_norm_0	999	57.1446	58.7451	42.657	57.1481	58.7566	42.5785
FD001_linear_rul_no_norm_0	Mean	57.1546	57.7103	42.54883	57.15287	57.71403	42.54783
FD001_linear_rul_1	42	21.7667	22.0821	24.1052	23.4334	23.4524	23.8384
FD001_linear_rul_1	1234	24.0107	23.4102	21.303	57.1724	56.6005	42.3133
FD001_linear_rul_1	999	23.3619	24.2572	21.8105	57.1482	58.7569	42.5766
FD001_linear_rul_1	Mean	23.04643	23.24983	22.40623	45.918	46.26993	36.24277
FD001_piecewise_rul_2	42	10.5972	10.4198	14.3786	41.3615	41.573	40.0648
FD001_piecewise_rul_2	1234	41.3774	41.6514	40.0138	41.3779	41.6514	40.0127
FD001_piecewise_rul_2	999	8.3692	8.4677	14.5316	10.8848	10.7415	13.8462

Figure 4. Effect of Different Seeds on RNN Training

To mitigate the effects of the Seed number in our best model, we chose to vary it along with three hyperparameters when optimising for the best model.

5. Evaluation

All models are evaluated using RMSE against the test set for FD001 for comparison. The chosen test set would be on the final feature engineered version of FD001 (RUL cap of 125, min-max normalisation, low-variance features removed). While intermediate evaluation uses RMSE for model selection, the ultimate objective is anomaly detection (classifying engines as healthy or in a danger zone) which is addressed in Section 7. Our first objective would be to determine which dataset resulted in the best performing models to choose a final comparison dataset. The FD001 datasets initially tested in sequence was:

1. Raw unnormalised linear RUL (no RUL cap) dataset
   (FD001_linear_rul_no_norm_0 / dc1_0)
2. Normalised linear RUL dataset
   (FD001_linear_rul_1 / dc1_1)
3. Normalised RUL-capped at 125 dataset
   (FD001_piecewise_rul_2 / dc1_2)
4. Normalised RUL-capped AND low variance features removed dataset
   (FD001_low_variance_1 / fe2_lv1)
5. Normalised RUL-capped, low varianced removed, s14 removed
   (FD001_drop_s14 / fe2_hc2_1)
6. Normalised RUL-capped, low varianced removed, s14 and s11 removed

   (FD001_drop_s14_s11 / fe2_hc2_2)
7. Normalised RUL-capped, low varianced removed, s14 and s12 removed

   (FD001_ drop_s14_s12 / fe2_hc2_3)

Before fine-tuning, each architecture stuck to baseline default hyperparameters such as a sequence length of 50 for the datasets we would use for training and testing and a learning rate of

0.001.

For hyperparameter tuning, each architecture would then vary the sequence length across [30,50,70], learning rates across [0.001, 0.0005, 0.0001].

5.1. RNN-Based Training Results and Selection of Final Comparison Dataset

To determine which dataset of the seven listed should be the one that we decide on for model comparisons, we first conduct our preliminary testing for all datasets cross multiple hyperparameters on the RNN architecture first since it is the simplest and quickest architecture for training runs.

More importantly, we acknowledge that results may often be skewed towards unlucky or lucky seeding from experimentation and thus, we run the 7 datasets across the 3 RNN model architectures across 3 seeds (42, 999, 1234 chosen arbitrarily) for training and testing. We take the average RMSE values across multiple seeds to average out the effect of lucky or unlucky seeds across models.

For further consistency, we also conduct hyperparameter tuning for the RNN architecture to simultaneously select the best RNN architecture model and the final comparison dataset through consistency. The hyperparameters that were tuned were sequence lengths of the dataset (30,50,70) and learning rate (1e-3, 5e-4, 1e-4). Additionally, we implement patience for early stopping on validation performance, so training halts when the validation RMSE does not improve for 5 consecutive epochs, which helps reduce overfitting and avoids unnecessary training time while retaining the best model checkpoint. The code for training the RNN-based models and plotting the loss curve is found at /RNN-based/tuning. Code to reproduce the best model's inference is found at /RNN-based/best_model_and_evaluation.

	Sequence Length	30									50									70								
	Models	RNN			GRU			LSTM			RNN			GRU			LSTM			RNN			GRU			LSTM		
Dataset	Seed	Train RMSE	Val RMSE	Test RMSE	Train RMSE	Val RMSE	Test RMSE	Train RMSE	Val RMSE	Test RMSE	Train RMSE	Val RMSE	Test RMSE	Train RMSE	Val RMSE	Test RMSE	Train RMSE	Val RMSE	Test RMSE	Train RMSE	Val RMSE	Test RMSE	Train RMSE	Val RMSE	Test RMSE	Train RMSE	Val RMSE	Test RMSE
FD001_linear_rul_no_norm_0	42	61.6896	61.1526	45.1676	61.6896	61.1526	45.1645	61.6897	61.1526	45.1612	57.1431	57.785	42.7109	57.1431	57.785	42.7017	57.1431	57.785	42.712	52.8603	53.7313	41.3093	52.8602	53.7313	41.3093	52.8603	53.7313	41.3076
FD001_linear_rul_no_norm_0	1234	61.7001	61.2308	44.8726	61.6923	61.2312	45.0603	61.6961	61.2305	44.9575	57.1802	56.6014	42.2586	57.1759	56.6008	42.2878	57.1674	56.6005	42.353	52.8612	53.8889	41.2885	52.8625	53.8931	41.2698	52.8616	53.8904	41.2817
FD001_linear_rul_no_norm_0	999	61.6904	63.3657	45.1289	61.6911	63.3676	45.1016	61.6904	63.3655	45.1311	57.1462	58.751	42.6154	57.1446	58.7451	42.657	57.1481	58.7566	42.5785	52.8608	52.678	41.2958	52.8612	52.679	41.2895	52.8616	52.68	41.2828
FD001_linear_rul_no_norm_0	Mean	61.69337	61.91637	45.05637	61.691	61.91713	45.1089	61.69207	61.9162	45.08377	57.1565	57.71247	42.5283	57.1546	57.7103	42.54883	57.15287	57.71403	42.54783	52.86077	53.43273	41.29743	52.8613	53.43447	41.28953	52.86117	53.4339	41.29073
FD001_linear_rul_1	42	61.6897	61.1526	45.164	26.8304	27.0541	25.2704	61.6897	61.1526	45.1603	57.143	57.785	42.7142	21.7667	22.0821	24.1052	23.4334	23.4524	23.8384	52.8603	53.7313	41.308	16.1069	16.714	21.8549	18.5449	19.2627	20.3475
FD001_linear_rul_1	1234	61.6991	61.2307	44.8926	30.8491	30.816	22.2192	61.6979	61.2305	44.9185	57.1771	56.6009	42.2794	24.0107	23.4102	21.303	57.1724	56.6005	42.3133	52.8612	53.8888	41.2889	17.4025	18.2398	22.0878	52.8617	53.8905	41.2811
FD001_linear_rul_1	999	61.6904	63.3657	45.1291	29.5754	30.8931	24.2392	61.6904	63.3656	45.1299	57.1462	58.751	42.6154	29.3619	24.2572	21.8109	57.1482	58.7569	42.5766	52.8608	52.678	41.296	16.1069	16.714	21.8549	52.8616	52.68	41.2828
FD001_linear_rul_1	Mean	61.69307	61.91633	45.0619	29.08487	29.58773	23.9096	61.69267	61.91623	45.06957	57.15543	57.7123	42.53633	23.04643	23.24983	22.40623	45.918	46.26993	36.24277	52.86077	53.4327	41.29763	16.53877	17.2226	21.93253	41.42273	41.9444	34.30373
FD001_piecewise_rul_2	42	41.7899	41.4892	40.4845	12.767	12.6758	13.3531	41.7901	41.9959	40.4784	41.3636	41.7048	40.0532	10.5972	10.4198	14.3786	41.3615	41.573	40.0648	40.4219	40.5343	39.5838	9.7996	9.9611	13.106	40.42	40.3524	39.585
FD001_piecewise_rul_2	1234	41.7918	41.7495	40.4488	13.0252	12.9612	13.1673	41.7912	41.7496	40.4568	41.3788	41.6514	40.0111	41.3774	41.6514	40.0138	41.3779	41.6514	40.0127	40.4185	40.5443	39.5869	7.8474	7.9305	14.6195	41.3779	41.6514	40.0127
FD001_piecewise_rul_2	999	41.79	41.9956	40.4822	12.8907	12.841	13.2384	41.7898	41.4892	40.4855	41.3605	41.5704	40.0728	8.3692	8.4677	14.5316	10.8848	10.7415	13.8462	40.4189	40.3492	39.5863	8.8034	9.1793	14.0345	40.4212	40.6343	39.5841
FD001_piecewise_rul_2	Mean	41.79057	41.74477	40.47183	12.8943	12.826	13.25293	41.79037	41.7448	40.47357	41.36763	41.6422	40.0457	20.1146	20.17963	22.97467	31.20807	31.32197	31.3079	40.41977	40.50527	39.58567	8.816867	9.023633	13.92	40.7397	40.87937	39.72727
FD001_low_variance_1	42	41.7901	41.4892	40.4782	12.8527	12.8443	13.2486	41.7896	41.4892	40.4966	41.3639	41.7048	40.0521	11.4437	11.265	13.1672	41.3624	41.7048	40.0596	40.4222	40.6343	39.5837	40.4217	40.6343	39.5839	40.4204	40.6343	39.5846
FD001_low_variance_1	1234	41.7921	41.7495	40.4449	12.6853	12.6209	13.2208	41.7916	41.7496	40.4511	41.3804	41.6515	40.0083	41.3822	41.6517	40.0053	41.3759	41.6514	40.0167	40.419	40.5461	39.5861	10.5828	10.4449	12.9159	40.4208	40.5514	39.5844
FD001_low_variance_1	999	41.7899	41.9956	40.4822	12.0829	12.2089	13.0339	41.7898	41.9954	40.4859	41.3602	41.5696	40.0755	10.9969	10.8444	13.9021	41.3593	41.5669	40.0853	40.42	40.3525	39.5849	9.9954	10.287	13.3189	40.4205	40.3538	39.5846
FD001_low_variance	Mean	41.7907	41.74477	40.46843	12.5403	12.55777	13.16777	41.79033	41.74473	40.47787	41.36817	41.64157	40.0453	21.27427	21.2537	22.3582	41.36587	41.64103	40.05393	40.4204	40.51967	39.5849	20.3333	20.4554	21.93957	40.42057	40.51317	39.58463
FD001_drop_s14	42	41.7901	41.4893	40.4775	12.958	12.9404	13.2429	41.7901	41.4892	40.4794	41.364	41.7049	40.0514	41.3631	41.7048	40.056	41.3632	41.7048	40.0554	40.422	40.6343	39.5838	40.4209	40.6343	39.5843	40.4217	40.6343	39.5839
FD001_drop_s14	1234	41.7918	41.7495	40.4494	12.8811	12.776	13.0796	41.7911	41.7497	40.4581	41.7918	41.7495	40.4494	41.3798	41.6515	40.0094	41.3782	41.6514	40.0122	40.4192	40.5467	39.5858	7.7618	7.823	15.2153	40.4198	40.5487	39.5813
FD001_drop_s14	999	41.7901	41.996	40.4776	12.6871	12.6727	13.6197	41.7903	41.9963	40.4731	41.3598	41.5687	40.0787	11.0468	10.9787	13.8588	41.3598	41.5686	40.0789	40.4188	40.349	39.5864	10.328	10.5992	14.3993	40.4199	40.3523	39.585
FD001_drop_s14	Mean	41.79067	41.74493	40.46817	12.84207	12.79637	13.31467	41.7905	41.74507	40.4702	41.5052	41.67437	40.19317	21.26323	31.445	31.30807	41.36767	41.6416	40.04883	40.42	40.51	39.58533	19.50357	19.6855	23.0663	40.42047	40.51177	39.58467
FD001_drop_s14_s11	42	41.79	41.4892	40.4813	12.8162	12.8253	13.8452	41.7901	41.4892	40.4787	41.3629	41.7048	40.0568	11.0441	11.0558	13.8076	41.3631	41.7048	40.056	40.4215	40.6343	39.584	8.7833	9.1637	15.0994	40.4219	40.6343	39.5838
FD001_drop_s14_s11	1234	41.7923	41.7495	40.4427	13.427	13.4004	13.3579	13.1043	13.1323	13.7215	41.3838	41.6519	40.0028	41.3856	41.6522	40.0002	41.3792	41.6514	40.0104	40.4204	40.5502	39.5846	8.5177	8.5668	15.8189	40.4211	40.5522	39.5842
FD001_drop_s14_s11	999	41.7906	41.9968	40.4672	13.2483	13.2378	13.4448	41.7899	41.9956	40.4827	41.3601	41.5695	40.0759	10.3394	10.3679	14.4213	41.3603	41.5699	40.0743	40.4196	40.3514	39.5853	40.4219	40.6343	39.5838	13.033	13.1202	13.6119
FD001_drop_s14_s11	Mean	41.79037	41.74517	40.46373	13.16383	13.1543	13.5493	32.2281	32.2057	31.56097	41.36833	41.64207	40.04517	20.92303	21.0253	22.74303	41.36753	41.64203	40.0469	40.4205	40.51157	39.58463	19.24157	19.4555	23.5007	31.292	31.43557	30.92663
FD001_drop_s14_s12	42	41.79	41.4892	40.4813	12.6442	12.6296	13.7207	41.7901	41.4892	40.4787	41.3629	41.7048	40.0567	11.051	10.9743	14.4012	41.363	41.7048	40.0561	40.4215	40.6343	39.584	9.3633	9.6567	14.4212	40.4219	40.6343	39.5838
FD001_drop_s14_s12	1234	41.7923	41.7495	40.4425	13.07	12.9399	13.3732	41.7927	41.7495	40.4389	41.3838	41.6519	40.0028	41.3855	41.6522	40.0003	41.3793	41.6514	40.0103	40.4204	40.5502	39.5846	6.955	7.0416	15.8255	40.4211	40.5522	39.5842
FD001_drop_s14_s12	999	41.7906	41.9968	40.4672	12.9265	13.032	13.3233	13.4686	13.3472	13.6658	41.3601	41.5695	40.0759	9.9807	10.0944	14.9045	11.292	11.1989	13.3753	40.4196	40.3514	39.5853	7.5719	7.926	16.2663	11.1156	11.2617	13.076
FD001_drop_s14_s12	Mean	41.79097	41.74517	40.46367	12.88023	12.86717	13.4724	32.35047	32.1953	31.5276	41.36833	41.64207	40.04513	20.80573	20.90697	23.102	31.34477	31.51833	31.14723	40.4205	40.51157	39.58463	7.9634	8.208167	15.50433	30.65267	30.81607	30.7483

Figure 5.1.1: Train, Val and Test RMSE for 7 datasets across [30,50,70] Sequence Lengths for RNN Architecture

	Learning Rate	1e(-3)									5e(-4)									1e(-4)								
	Models	RNN			GRU			LSTM			RNN			GRU			LSTM			RNN			GRU			LSTM		
Dataset	Seed	Train RMSE	Val RMSE	Test RMSE	Train RMSE	Val RMSE	Test RMSE	Train RMSE	Val RMSE	Test RMSE	Train RMSE	Val RMSE	Test RMSE	Train RMSE	Val RMSE	Test RMSE	Train RMSE	Val RMSE	Test RMSE	Train RMSE	Val RMSE	Test RMSE	Train RMSE	Val RMSE	Test RMSE	Train RMSE	Val RMSE	Test RMSE
FD001_linear_rul_no_norm_0	42	57.1431	57.785	42.7109	57.1433	57.785	42.7017	57.1431	57.785	42.712	57.1429	57.785	42.7182	57.1431	57.785	42.711	57.1431	57.785	42.7089	66.6768	67.059	46.6265	66.8721	67.2513	46.8069	48.5668	48.5582	45.5609
FD001_linear_rul_no_norm_0	1234	57.1802	56.6014	42.2586	57.1759	56.6008	42.2878	57.1674	56.6005	42.353	57.1703	56.6005	42.3299	57.1663	56.6006	42.3622	57.1628	56.601	42.3932	66.8887	65.5154	46.8223	66.6484	65.2828	46.6003	48.8545	48.4543	45.8252
FD001_linear_rul_no_norm_0	999	57.1462	58.751	42.6154	57.1446	58.7451	42.657	57.1481	58.7566	42.5785	57.1467	58.7524	42.606	57.1465	58.7518	42.6101	57.1466	58.7521	42.6079	67.0376	69.2397	46.9606	66.9171	69.1175	46.8486	48.5251	49.0224	45.5226
FD001_linear_rul_no_norm_0	Mean	57.1565	57.71247	42.5283	57.1546	57.7103	42.54883	57.15287	57.71403	42.54783	57.1533	57.71263	42.55137	57.15197	57.71247	42.5611	57.15083	57.7127	42.57	66.8677	67.27137	46.80313	66.81253	67.2177	46.75193	48.6488	48.68163	45.63623
FD001_linear_rul_1	42	57.143	57.785	42.7142	21.7667	22.0821	24.1052	23.4334	23.4524	23.8384	57.1679	56.6005	42.349	25.4291	25.3844	19.3022	27.057	26.9486	19.1054	66.737	65.3685	46.6819	66.7332	67.1145	46.6784	66.6211	67.0042	46.5751
FD001_linear_rul_1	1234	57.1771	56.6009	42.2794	24.0107	23.4102	21.303	57.1724	56.6005	42.3133	57.1467	58.7524	42.6061	57.1661	56.6006	42.3643	57.1667	56.6006	42.3588	67.0235	69.2255	46.9475	66.619	65.2542	46.5731	66.7764	65.4066	46.7183
FD001_linear_rul_1	999	57.1462	58.751	42.6154	23.3619	24.2572	21.8109	57.1482	58.7569	42.5766	57.1429	57.785	42.7186	26.2653	27.284	18.9617	30.0089	31.4468	16.8649	66.5946	66.9781	46.5508	66.8729	69.0727	46.8076	67.0834	69.2662	47.0032
FD001_linear_rul_1	Mean	57.15543	57.7123	42.53633	23.04643	23.24983	22.40623	45.918	46.26993	36.24277	57.1525	57.71263	42.5579	36.29683	36.423	26.87697	38.07753	38.332	26.1097	66.78503	67.1507	46.72673	66.7417	67.14713	46.68637	66.82697	67.23233	46.76553
FD001_piecewise_rul_2	42	41.3636	41.7048	40.0532	10.5972	10.4198	14.3786	41.3615	41.573	40.0648	41.3771	41.6514	40.0143	11.0388	10.8013	13.556	11.7299	11.5344	14.1392	48.6806	48.2974	45.6653	48.671	48.66	45.6565	48.5589	48.5504	45.5536
FD001_piecewise_rul_2	1234	41.3788	41.6514	40.0111	41.3774	41.6514	40.0138	41.3779	41.6514	40.0127	41.3625	41.575	40.0592	41.3766	41.6513	40.0153	41.3761	41.6514	40.0164	48.9609	49.4638	45.9229	48.8159	48.4273	45.7896	48.7188	48.3341	45.7005
FD001_piecewise_rul_2	999	41.3605	41.5704	40.0728	8.3692	8.4677	14.5316	10.8848	10.7415	13.8462	41.3622	41.7048	40.0609	11.221	11.2002	12.7336	41.3618	41.5735	40.0633	48.5327	48.5249	45.5297	48.8069	49.3081	45.7814	49.0093	49.5133	45.9679
FD001_piecewise_rul_2	Mean	41.36763	41.6422	40.0457	20.1146	20.17963	22.97467	31.20807	31.32197	31.3079	41.36727	41.64373	40.0448	21.21213	21.2176	22.10163	31.48927	31.58643	31.4063	48.7246	48.76203	45.70597	48.7646	48.75847	45.74257	48.76233	48.79927	45.74067
FD001_low_variance_1	42	41.3639	41.7048	40.0521	11.4437	11.265	13.1672	41.3624	41.7048	40.0596	41.3797	41.6515	40.0095	11.2157	10.9607	13.2962	41.3639	41.7049	40.0521	48.8281	48.4391	45.801	48.3015	48.2991	45.3179	48.2037	48.2035	45.2286
FD001_low_variance_1	1234	41.3804	41.6515	40.0083	41.3822	41.6517	40.0053	41.3759	41.6514	40.0167	41.361	41.5718	40.0683	41.3783	41.6514	40.012	41.3727	41.6516	40.0237	48.6196	49.1182	45.6093	48.561	48.1826	45.5555	48.6364	49.1353	45.6248
FD001_low_variance_1	999	41.3602	41.5696	40.0755	10.9969	10.8444	13.9021	41.3593	41.5669	40.0853	41.3627	41.7048	40.0576	10.9726	10.938	13.447	41.3609	41.5715	40.0692	49.0956	49.0751	46.0475	48.5272	49.0246	45.5246	48.6364	49.1353	45.6248
FD001_low_variance	Mean	41.36817	41.64157	40.0453	21.27427	21.2537	22.3582	41.36587	41.64103	40.05393	41.3676	41.6427	40.04513	21.18883	21.18337	22.25173	41.36583	41.64267	40.04833	48.84777	48.87747	45.81927	48.46323	48.5021	45.466	48.49217	48.8247	45.45273
FD001_drop_s14	42	41.364	41.7049	40.0514	41.3631	41.7048	40.056	41.3632	41.7048	40.0554	41.3784	41.6514	40.0118	41.3626	41.7048	40.0584	41.3616	41.7049	40.0641	48.6136	48.6039	45.6039	48.5299	48.5221	45.527	48.5655	48.557	45.5597
FD001_drop_s14	1234	41.7918	41.7495	40.4494	41.3798	41.6515	40.0094	41.3782	41.6514	40.0122	41.3613	41.5725	40.0663	41.4268	41.4291	12.6184	41.3751	41.6514	40.0183	48.8581	48.4678	45.8286	48.7348	48.3494	45.7152	48.4432	48.0697	45.4476
FD001_drop_s14	999	41.3598	41.5687	40.0787	11.0468	10.9787	13.8588	41.3598	41.5686	40.0789	41.3623	41.7048	40.0599	41.3619	41.5738	40.0625	41.3615	41.5729	40.0649	48.559	49.0568	45.5537	48.4208	48.9167	45.4271	48.5232	49.0205	45.5209
FD001_drop_s14	Mean	41.5052	41.67437	40.19317	21.26323	31.445	31.30807	41.36707	41.6416	40.04883	41.36733	41.6429	40.046	31.38377	31.56823	30.9131	41.36607	41.64307	40.0491	48.6789	48.7095	45.66207	48.56183	48.59607	45.55643	48.51063	48.54907	45.5094
FD001_drop_s14_s11	42	41.3629	41.7048	40.0568	11.0441	11.0558	13.8076	41.3631	41.7048	40.056	41.377	41.6513	40.0145	11.2891	11.0509	13.9981	41.3776	41.6514	40.0134	48.7148	48.3302	45.6968	48.794	48.7802	45.7696	48.5635	48.555	45.5579
FD001_drop_s14_s11	1234	41.3838	41.6519	40.0028	41.3856	41.6522	40.0002	41.3792	41.6514	40.0104	41.3614	41.5727	40.0655	11.3016	11.245	13.7835	41.3776	41.6514	40.0134	48.5398	49.0374	45.5362	48.9401	48.5466	45.9041	48.8521	48.462	45.823
FD001_drop_s14_s11	999	41.3601	41.5695	40.0759	10.3394	10.3679	14.4213	41.3603	41.5699	40.0743	41.3629	41.7048	40.0566	11.313	11.285	13.4175	41.3614	41.5726	40.066	48.676	48.6649	45.6611	48.5548	49.0525	45.5499	48.5553	49.053	45.5503
FD001_drop_s14_s11	Mean	41.36893	41.64207	40.04513	20.92303	21.0253	22.74303	41.36753	41.64203	40.0469	41.3671	41.64293	40.04553	11.30123	11.19983	13.73303	41.3722	41.62513	40.03093	48.64353	48.6775	45.63137	48.76297	48.7931	45.7412	48.65697	48.69	45.64373
FD001_drop_s14_s12	42	41.3629	41.7048	40.0567	11.051	10.9743	14.4012	41.363	41.7048	40.0561	41.377	41.6513	40.0145	11.2554	11.0786	13.1946	41.3617	41.7049	40.0638	48.7124	48.328	45.6946	48.7901	48.7764	45.766	48.5668	48.5582	45.5609
FD001_drop_s14_s12	1234	41.3838	41.6519	40.0028	41.3855	41.6522	40.0003	41.3793	41.6514	40.0103	41.3614	41.5727	40.0655	11.6906	11.6888	12.6516	41.3776	41.6514	40.0134	48.5416	49.0391	45.5377	48.937	48.5436	45.9012	48.8545	48.4543	45.8252
FD001_drop_s14_s12	999	41.3601	41.5695	40.0759	9.9807	10.0944	14.9045	11.292	11.1989	13.3753	41.363	41.7048	40.0566	13.9072	14.0156	13.0064	13.1307	13.1486	13.678	48.6862	48.6749	45.6705	48.5556	49.0533	45.5506	48.5251	49.0224	45.5226
FD001_drop_s14_s12	Mean	41.36893	41.64207	40.04513	20.80573	20.90697	23.102	31.34477	31.51833	31.14723	41.36713	41.64293	40.0455	12.2844	12.261	12.95087	31.95667	32.1683	31.25173	48.64673	48.68067	45.63427	48.7609	48.7911	45.73927	48.6488	48.68163	45.63623

Figure 5.1.2: Train, Val and Test RMSE for 7 datasets across [1e-3, 5e-4, 1e-4] Learning Rates for RNN Architecture

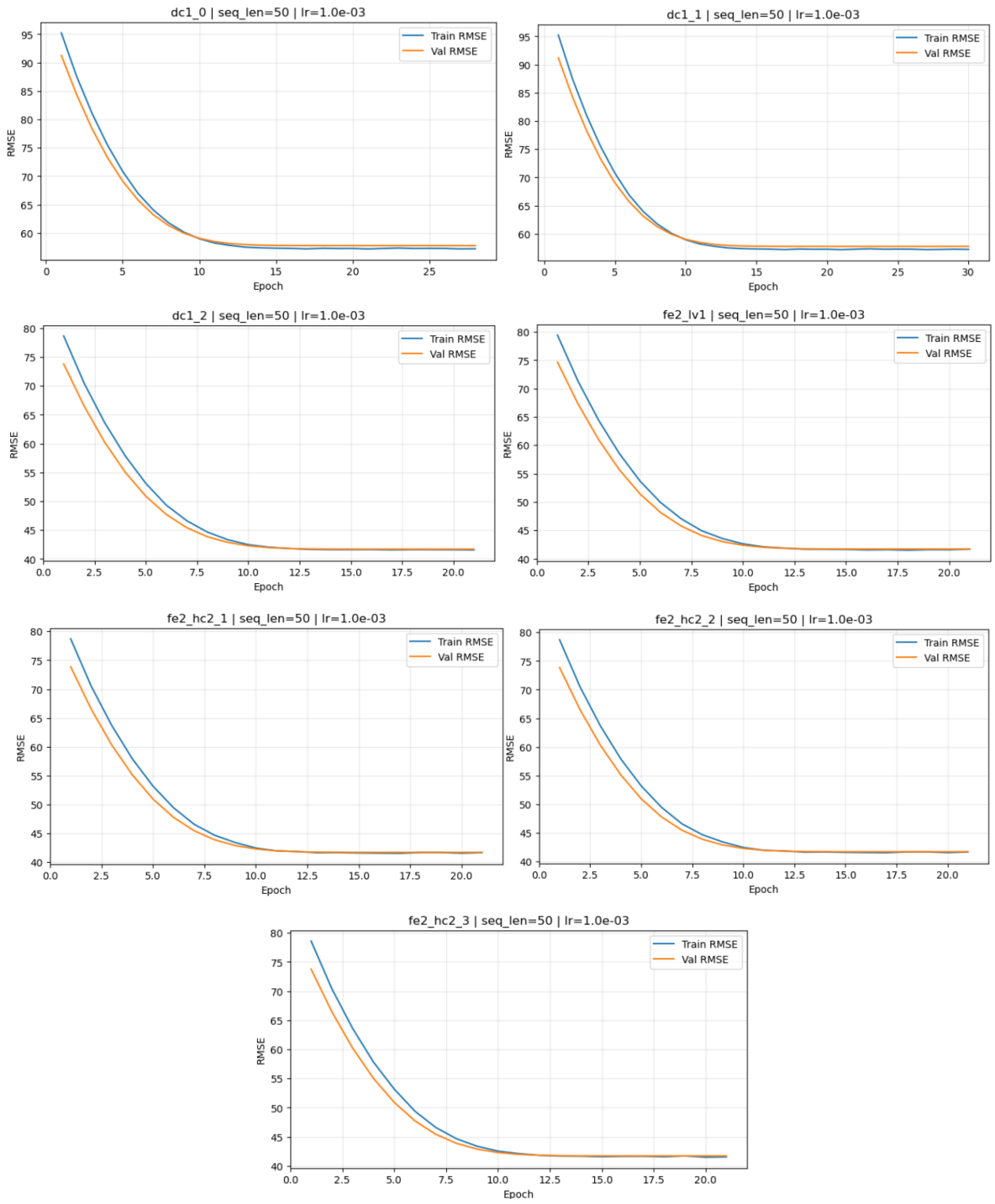


Figure 5.1.3: Loss Curves for RNN Architecture (Seed 42) across 7 datasets at SEQ_LEN = 50, BATCH_SIZE = 64, EPOCHS = 50 and LEARNING_RATE = 1e-3 (SEQ_LEN=30, SEQ_LEN =70 not shown for brevity)

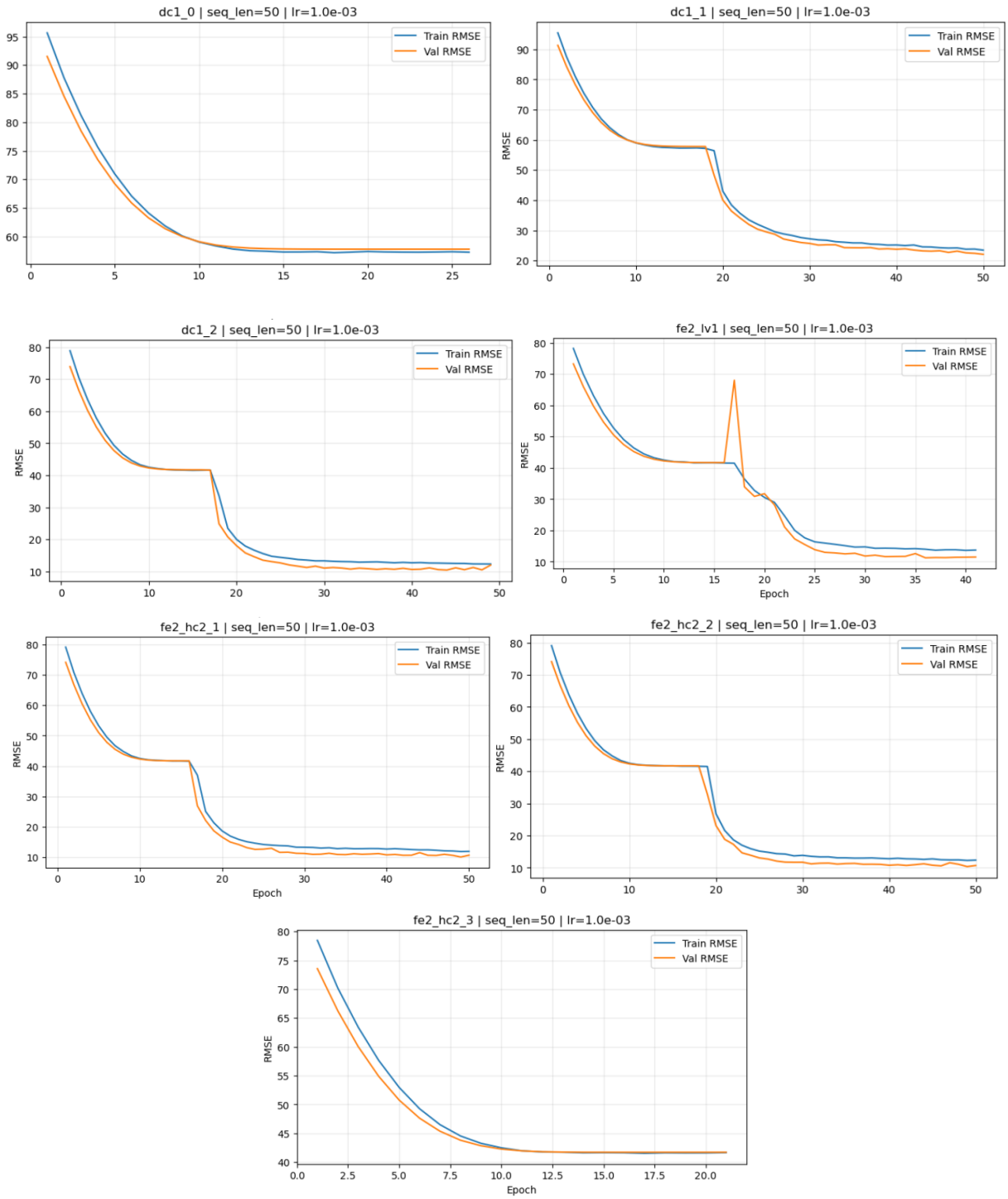
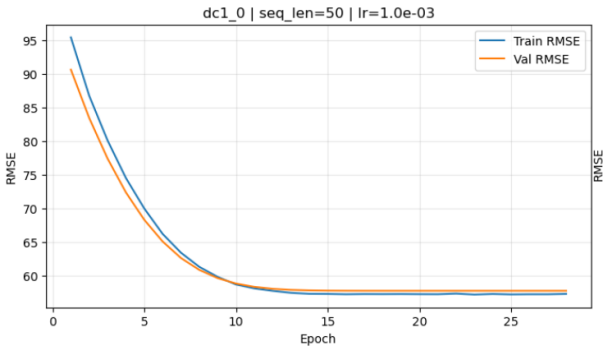
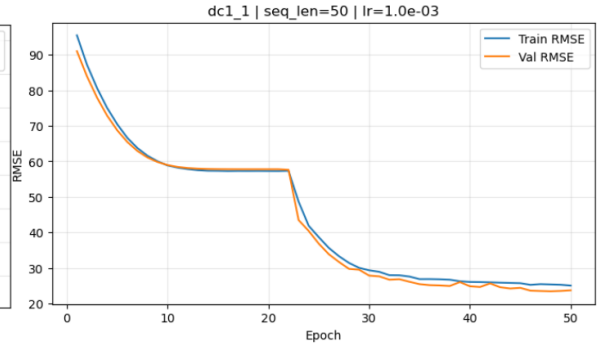
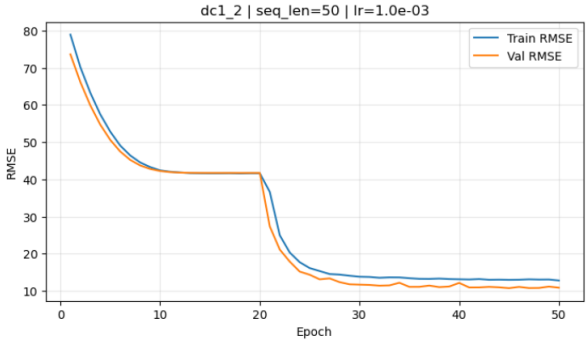
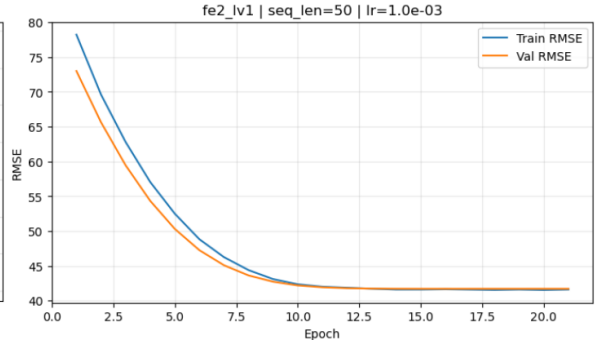


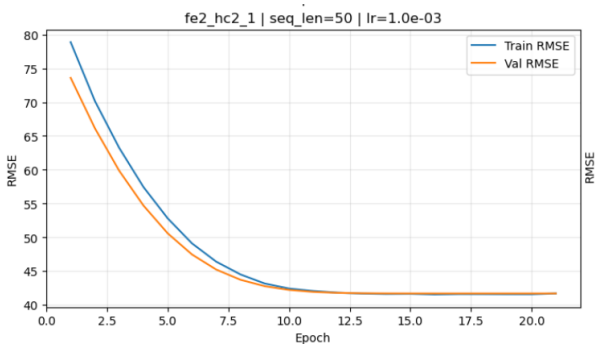
Figure 5.1.4: Loss Curves for GRU Architecture (Seed 42) across 7 datasets at SEQ_LEN = 50, BATCH_SIZE = 64, EPOCHS = 50 and LEARNING_RATE = 1e-3 (SEQ_LEN=30, SEQ_LEN =70 not shown for brevity)

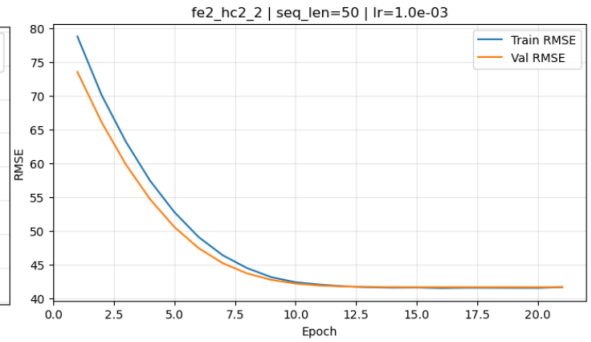

dc1_0 | seq_len=50 | lr=1.0e-03
Train RMSE
Val RMSE
RMSE
Epoch


dc1_1 | seq_len=50 | lr=1.0e-03
Train RMSE
Val RMSE
RMSE
Epoch


dc1_2 | seq_len=50 | lr=1.0e-03
Train RMSE
Val RMSE
RMSE
Epoch


fe2_lv1 | seq_len=50 | lr=1.0e-03
Train RMSE
Val RMSE
RMSE
Epoch


fe2_hc2_1 | seq_len=50 | lr=1.0e-03
Train RMSE
Val RMSE
RMSE
Epoch


fe2_hc2_2 | seq_len=50 | lr=1.0e-03
Train RMSE
Val RMSE
RMSE
Epoch

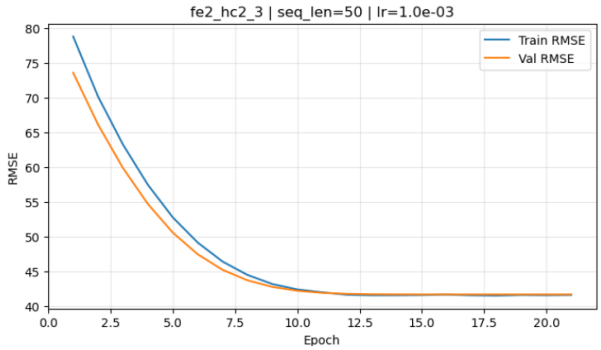

fe2_hc2_3 | seq_len=50 | lr=1.0e-03
Train RMSE
Val RMSE
RMSE
Epoch

Figure 5.1.5: Loss Curves for LSTM Architecture (Seed 42) across 7 datasets at SEQ_LEN = 50, BATCH_SIZE = 64, EPOCHS = 50 and LEARNING_RATE = 1e-3 (SEQ_LEN=30, SEQ_LEN =70 not shown for brevity)

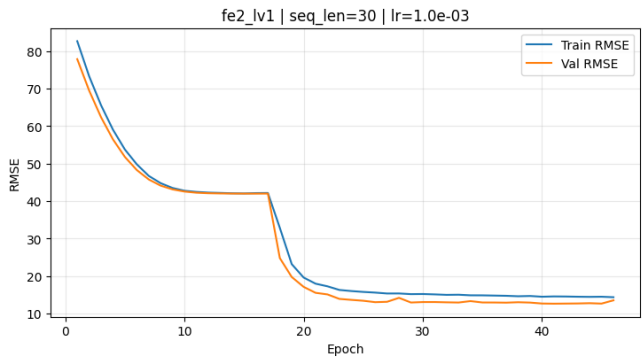


Figure 5.1.6: Loss Curve for Final Representative Model for RNN Architecture (GRU, SEQ_LEN = 30, LEARNING_RATE = 1e-3) for the low-variance dataset

From our analysis using solely the RNN architecture, we can determine which datasets as well as which hyperparameters for the RNN lead to the best training and validation performance. More importantly, for the rest of this comparison, we can use the above figures to check for consistency of results to standardise the dataset we would be using moving forward when comparing the different model architectures.

From the figures above, most of the minimum validation RMSEs occurred with the GRUs using the dataset that only had low variance features removed. Through this exercise, we can determine the finalised dataset for comparison across the different architectures which is the normalised RUL-capped AND only low variance features removed FD001 dataset (found in data/processed-nasa-data/feature_engineering_2/low_variance_1).

A notable observation from Figure 5.1.4 is the sharp validation RMSE spike visible in the GRU loss curve for the low-variance feature-removed dataset (fe2_lv1) at approximately epoch 17,

where validation RMSE jumps from ~42 to ~68 before sharply recovering and continuing its descent to a final value of ~12. Critically, the training RMSE does not spike at the same epoch and it continues its smooth decline. The spike is specific to the GRU on this dataset; it is not observed in the RNN or LSTM curves for the same dataset.

We attribute this to the GRU's reset and update gate dynamics interacting with the reduced feature space of fe2_lv1. After low-variance features are removed, the remaining features carry denser, more concentrated degradation signals. In the early training phase (epochs 1–15), the GRU's gates converge on a hidden state representation optimised for the training engine distribution. However, around epoch 17, the gate weights appear to undergo a rapid phase transition (i.e. a sudden reorganisation of what information the update gate retains versus discards across time steps) which temporarily degrades generalisation to the held-out validation engines before the model settles into a better basin. This phase transition does not manifest in the vanilla RNN because the RNN has no gating mechanism to reorganise, and does not manifest as strongly in the LSTM because the LSTM's separate cell state provides an additional memory pathway that stabilises the transition.

Practically, this spike does not affect our final results because our checkpointing mechanism saves only the model with the lowest validation RMSE seen so far. The checkpoint captured before the spike is therefore retained, and the model correctly recovers to its best performance after the spike resolves. This also validates the importance of epoch-level checkpointing rather than simply saving the final epoch's weights.

5.2. Transformer Training Results

The results of Transformer training with constants of SEQ_LEN = 50, BATCH_SIZE = 64, EPOCHS = 50 and LEARNING_RATE = 1e-3 for dataset exploration can be found in the table below.

Dataset	Seed	Train RMSE	Val RMSE	Test RMSE
FD001_linear_rul_no_norm_0	42	58.54706821	56.02898791	61.7254982
FD001_linear_rul_no_norm_0	1234	58.17394754	56.02924084	61.84631199
FD001_linear_rul_no_norm_0	999	58.1233467	56.02896805	61.74715727
FD001_linear_rul_no_norm_0	Mean	58.28145415	56.0290656	61.77298915
FD001_linear_rul_1	42	26.31345529	24.10437263	33.96547199
FD001_linear_rul_1	1234	27.17489287	23.78853537	31.91223587
FD001_linear_rul_1	999	24.97198596	22.26687075	34.00567469
FD001_linear_rul_1	Mean	26.15344471	23.38659291	33.29446085
FD001_piecewise_rul_2	42	8.775126323	12.31746412	14.43558323
FD001_piecewise_rul_2	1234	12.88504937	12.45414731	13.75048425
FD001_piecewise_rul_2	999	13.94760526	12.7632214	12.72602739
FD001_piecewise_rul_2	Mean	11.86926032	12.51161094	13.63736496
FD001_low_variance_1	42	13.57910535	12.36410914	13.47903358
FD001_low_variance_1	1234	12.62977576	12.8310931	14.22410427
FD001_low_variance_1	999	14.32355593	12.97299038	13.2554822
FD001_low_variance_1	Mean	13.51081235	12.72273087	13.65287335

Fig 5.2.1. Transformer Results Across 4 Datasets with Varying Seed

As mentioned in Section 5.1, the Transformer model then underwent hyperparameter optimisation using the finalised low_variance_1 dataset. The hyperparameters search space includes SEEDS = [42, 1234, 999], SEQ_LENS = [30, 50, 70, 100], LEARNING_RATES = [1e-4, 5e-4, 1e-3], NUM_LAYERS = [1, 2, 4].

The results of the hyperparameter optimisation can be found in /Transformer/hopt_lowvar_results.csv. The best model's loss curve logged every 5 epochs can be found below. The code for training the Transformer model and plotting the loss curve is found at /Transformer/transformer-with-opt.ipynb. Code to reproduce the best model's inference is found at /Transformer/recreate-transformer-inference.ipynb.

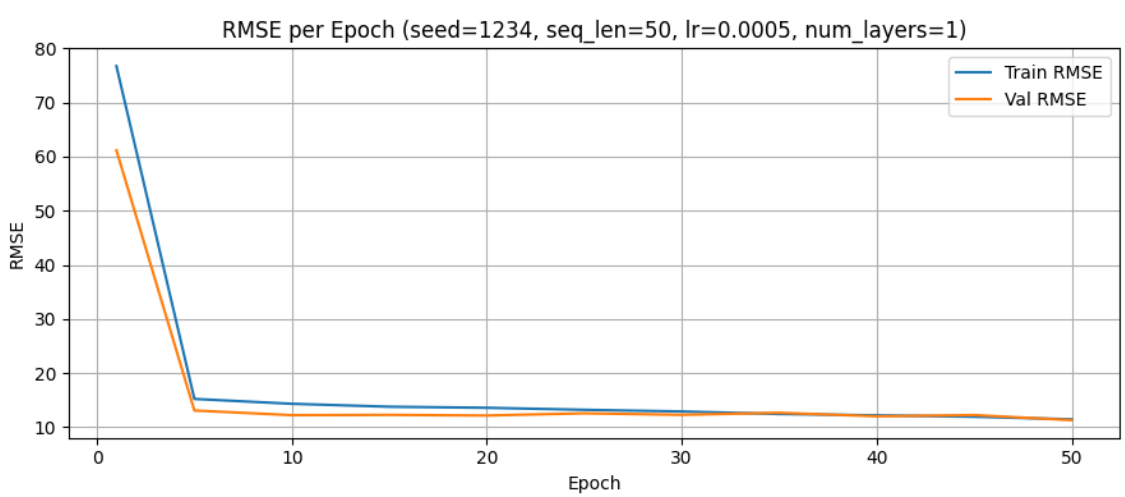


Fig 5.2.2. Loss Curve for Final Representative Model for Transformer Architecture for the low-variance dataset

5.3. Hybrid-Transformer Training Results

The results of Hybrid-Transformer training with constants of SEQ_LEN = 50, BATCH_SIZE = 64, EPOCHS = 50 and LEARNING_RATE = 1e-3 can be found in the table below.

		LSTM-Transformer			GRU-Transformer			CNN-Transformer		
Dataset	**Seed**	**Train RMSE**	**Val RMSE**	**Test RMSE**	**Train RMSE**	**Val RMSE**	**Test RMSE**	**Train RMSE**	**Val RMSE**	**Test RMSE**
FD001_linear_rul_no_norm_0	42	57.0253	57.8391	60.0582	57.2041	58.0349	58.7123	95.0117	95.6328	70.4581
FD001_linear_rul_no_norm_0	1234	57.2727	56.6201	61.3893	57.272	56.6216	61.3579	94.9576	95.1022	91.132
FD001_linear_rul_no_norm_0	999	56.9847	58.7896	58.8237	56.7989	58.7031	60.1096	74.4036	76.6299	93.2428
FD001_linear_rul_no_norm_0	Mean	57.0942333	57.7496	60.0904	57.09166667	57.78653333	60.0599333	88.1243	89.12163333	84.9443
FD001_linear_rul_1	42	9.421	9.794	41.8113	7.625	8.1799	37.2295	6.5731	6.8227	39.9406
FD001_linear_rul_1	1234	14.7703	14.887	42.1203	7.0735	7.4115	39.5189	9.8431	9.8325	37.3936
FD001_linear_rul_1	999	13.4687	14.7488	37.9963	7.9547	9.0622	39.5042	7.3639	7.761	38.4943
FD001_linear_rul_1	Mean	12.5533333	13.1432667	40.64263333	7.551066667	8.217866667	38.7508667	7.9267	8.138733333	38.6095
FD001_piecewise_rul_2	42	5.9262	6.1012	22.9607	5.0369	5.5184	22.7324	6.0172	6.0666	25.0446
FD001_piecewise_rul_2	1234	7.4283	7.5048	23.5339	6.1569	6.353	**21.1493**	6.4818	6.5687	23.2652
FD001_piecewise_rul_2	999	6.8432	7.2057	22.4157	6.7639	7.236	22.035	7.1842	7.2914	24.5674
FD001_piecewise_rul_2	Mean	6.73256667	6.93723333	22.9701	5.9859	6.369133333	21.9722333	6.561066667	6.642233333	24.2924
FD001_low_variance_1	42	6.0377	6.4179	24.1528	5.7156	6.1042	22.7581	5.5036	5.5055	24.5802
FD001_low_variance_1	1234	7.879	8.1091	22.4813	6.0697	6.2556	21.4482	5.0284	5.1084	25.0681
FD001_low_variance_1	999	10.2546	11.1323	22.7723	5.1127	5.5614	22.1624	4.9074	4.944	**22.4067**
FD001_low_variance	Mean	8.0571	8.5531	**23.13546667**	5.632666667	5.973733333	**22.1229**	5.146466667	5.185966667	**24.01833333**

Fig 5.3.1 Hybrid-Transformer Training Results

Once the best dataset has been selected, we proceed to hyperparameter finetuning for the three models in which the hyperparameters search space includes

SEEDS = [42, 1234, 999],

SEQ_LENS = [30, 50, 100],

LEARNING_RATES = [1e-4, 5e-4, 1e-3],

D_MODEL = [32, 64, 128].

The code for training the Hybrid models and plotting the loss curves for all datasets is found at Hybrid/FD001_hybrid_transformers_FE_AWS_SagemakerAI.ipynb, and

Code to reproduce the best model's inference is found at Hybrid/FD001_hyperparamter_tunning.ipynb.

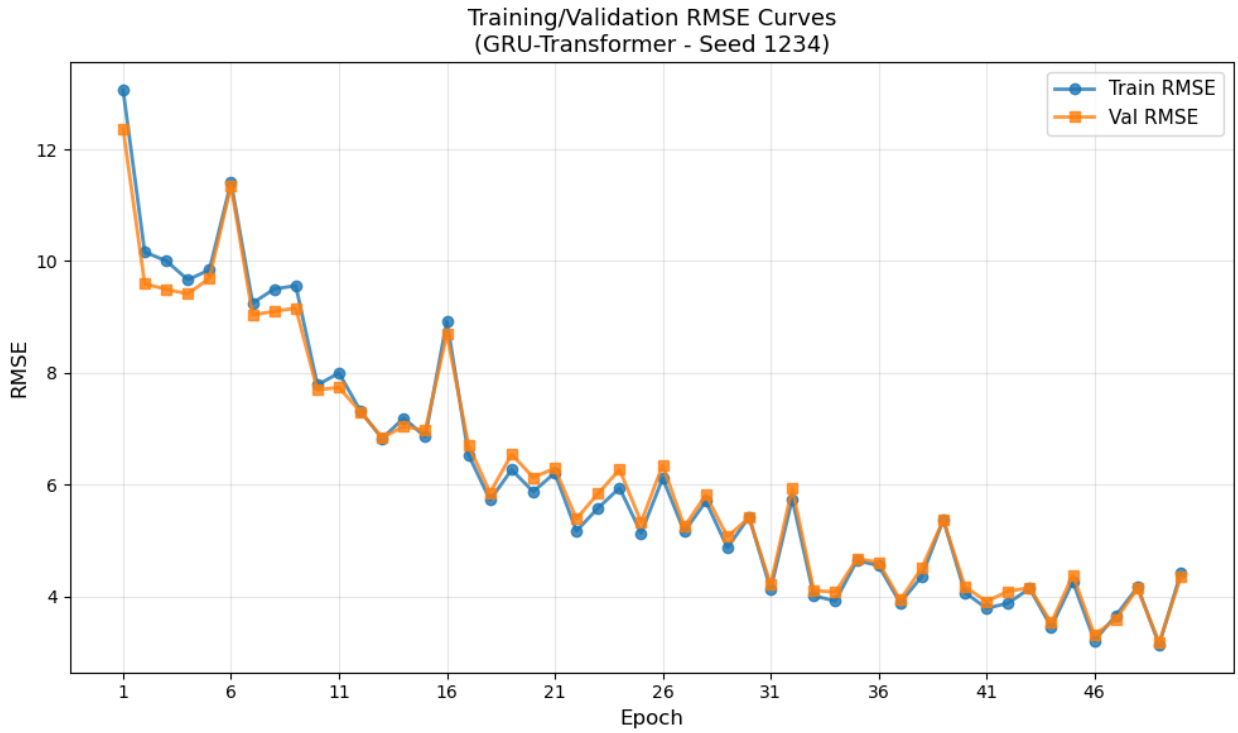


Fig 5.3.2. Loss Curve for The Optimized GRU-Transformer Model for the low-variance dataset

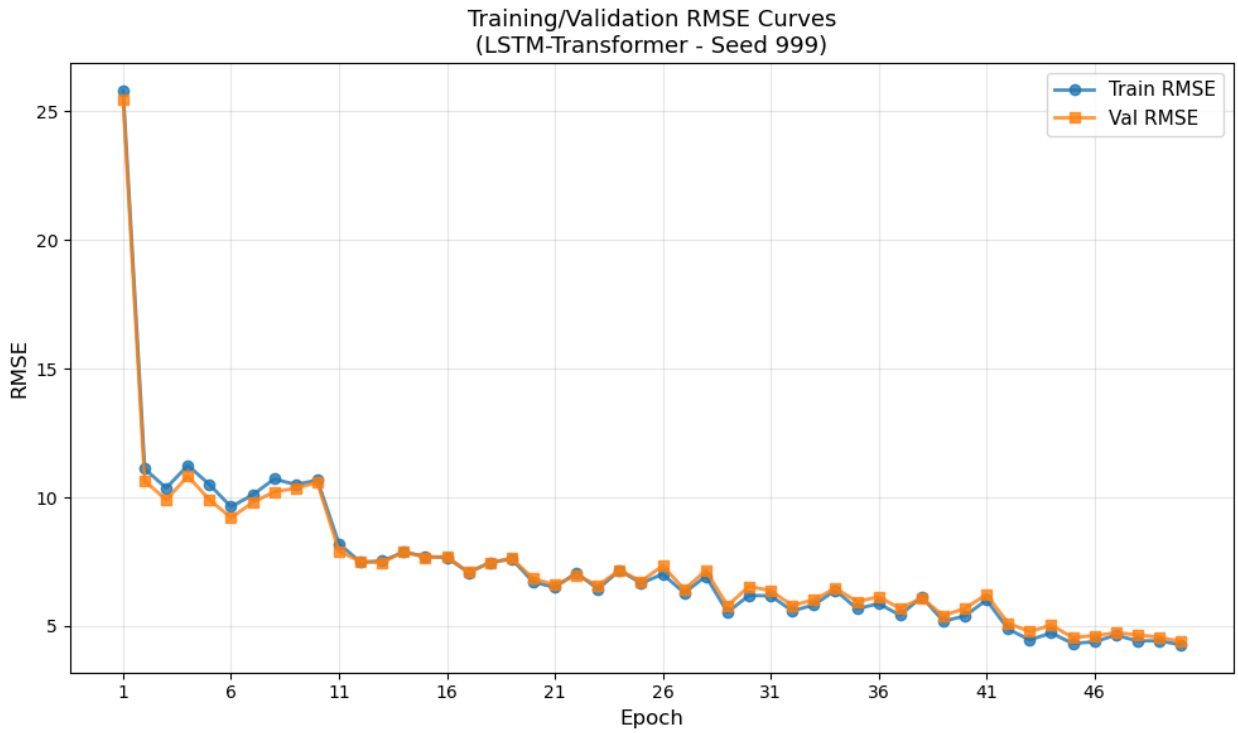


Fig 5.3.3. Loss Curve for The Optimized LSTM-Transformer Model for the low-variance dataset

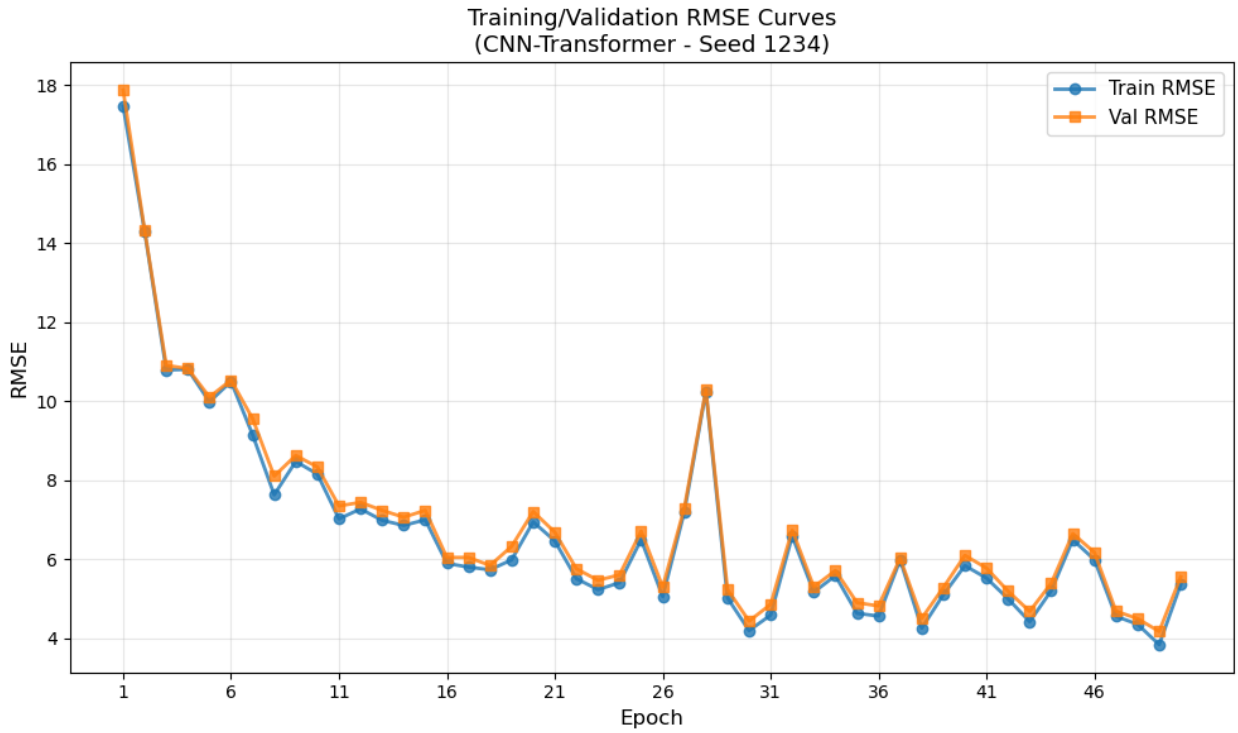


Fig 5.3.4. Loss Curve for The Optimized CNN-Transformer Model for the low-variance dataset

5.4. Best Results for Each Model (Before Finetuning)

RNN-based			Transformer	Hybrid		
RNN	**GRU**	**LSTM**	**'RULTransformer'**	**CNN-Trans**	**GRU-Trans**	**LSTM-Trans**
40.06	22.36	40.05	13.21	24.02	22.12	23.13

Fig 5.4: Best Mean RMSE Values for Each Architecture Pre-finetuning

5.5. Best Results for Each Model (After Finetuning)

RNN-based			Transformer	Hybrid		
RNN	**GRU**	**LSTM**	**'RULTransformer'**	**CNN-Trans**	**GRU-Trans**	**LSTM-Trans**
38.58	13.17	39.58	12.41	16.40	14.74	14.65

Fig 5.5: Best Mean RMSE Values for Each Architecture Post-finetuning

5.6. Benchmarking with SOTA

Model/Reference	**Source**	**RSME for FD001 dataset**
RNN	This work	38.58
GRU	This work	13.17
LSTM	This work	39.58
Transformer	*This work*	***12.41***
CNN-Transformer	This work	16.40
GRU-Transformer	This work	14.74
LSTM-Transformer	This work	14.65
LSTM-MLP-OR	J. P. Noot, M. Martin, and E. Birmele. (Jul. 2025).	14.24
LSTM-MLP-ORE	J. P. Noot, M. Martin, and E. Birmele. (Jul. 2025).	13.20
GCU-Transformer	Y. Mo, Q. Wu, X. Li, and B. Huang, (Oct. 2021).	11.27
AGCTE	M. Ma, Z. Wang, and Z. Zhong. (Apr. 2024)	12.46
DSFormer	J. Li et al., (2024).	10.77

Fig 5.6: Summary of SOTA Results

6. Intepretation of Results

The standalone Transformer performs best among the tested models, achieving an RMSE score of 12.41 after hyperparameter fine-tuning. This demonstrates that pure self-attention mechanisms are highly effective for CMAPSS RUL prediction. While hybrid models attempt to combine sequential memory with global attention, the standalone Transformer proved to be more effective. Because it directly captures long-range dependencies without relying on a recurrent front-end, it achieves strong predictive power while avoiding unnecessary architectural complexity.

This aligns with our observations regarding model capacity and dataset size. The CMAPSS dataset, particularly the FD001 subset, is modest in scale with only ~100 engines. Adding recurrent feature extractors (like GRUs or LSTMs) to a Transformer block increases the overall parameter count, making those hybrid models more susceptible to overfitting on limited data compared to the streamlined self-attention approach.

Furthermore, when evaluating the recurrent components, GRUs generally outperformed LSTMs in both the pure RNN-based and hybrid architectures. As mentioned in Section 3.5.1.3, the LSTM has a greater number of parameters to be trained as a result of its dedicated cell state and more complex gating mechanisms. In the context of the limited FD001 dataset, the LSTM's heavier parameter footprint made it significantly more prone to overfitting, which led to its underperformance across all tested architectures relative to the simpler GRU.

7. Anomaly Detection from Regression to Classification

Having established a strong regression foundation for RUL prediction, we now address the core objective of this project: anomaly detection under class imbalance. Providing a continuous numerical RUL is not always the most actionable format for maintenance teams, who typically operate on binary decision thresholds (e.g., "Is this engine safe to fly, or does it need immediate inspection?"). Rather than training a standalone classifier from scratch, we leverage the regression model's learned health representation directly by applying a threshold to its output to produce binary anomaly labels. This approach tests whether a well-trained regression model implicitly learns a latent space that is discriminative enough to support anomaly detection without any classification-specific training signal.

We explored converting the Transformer regression output into an "Engine-Level" classification (predicting if a new engine would suffer a premature failure based solely on its first 50 cycles). However, this highlighted a fundamental constraint of the NASA C-MAPSS simulated dataset: all engines begin in a perfectly healthy state, and degradation signatures do not numerically manifest until the final ~125 cycles (corresponding with our choice to cap RUL at 125). Therefore, attempting to classify an engine's total lifespan early in its operational life was not physically or mathematically viable.

To align the model's capabilities with real-world predictive maintenance logic, we transitioned to a **Current State Classification** approach. Instead of forecasting total lifespan from early data, the model evaluates the engine's most recent cycle window to determine its current operational health. The size of the window is taken from the ideal seq_len obtained from fine-tuning.

Based on standard predictive maintenance practices for the CMAPSS dataset, we established a critical RUL threshold of **30 cycles** (Sharma, 2025). This represents a realistic operational warning window, giving maintenance crews ample time to ground an aircraft before a catastrophic failure.

Crucially, establishing this 30-cycle threshold frames the problem as an imbalanced anomaly detection task, satisfying the project's core requirement. By defining any sensor reading with an **RUL <= 30** as 'Danger Zone' and **RUL > 30** as 'Healthy', the dataset naturally exhibits a heavy class imbalance. For example, in the FD001 training set (20,631 total rows across 100 engines), each engine only spends its final 31 cycles in the anomalous state. This results in exactly 3,100 'Danger Zone' samples against 17,531 'Healthy' samples. This gives an approximate **85% Healthy to 15% Anomalous** split.

The task is now a binary classification: Healthy (RUL > 30) or Danger Zone (RUL <= 30). This was achieved efficiently via post-processing. We retained the optimized regression weights and applied the threshold directly to the inference output.

For test set evaluation, only the final known cycle window per engine (i.e. a single sequence of the last 70 cycles) is constructed for each of the 100 test engines, and the model produces one classification label per engine.

No retraining, fine-tuning, or class-weighted loss is applied at the classification stage. This is intentional as it tests whether the regression model's learned latent representation of engine health is discriminative enough to support binary anomaly detection purely through threshold inference, without any classification-specific training signal.

This transition bridged the gap between raw machine learning metrics and practical operational utility. The resulting system effectively translates the regression tracking into a clear, reliable early-warning alert system for impending engine failure.

Classification Results (Threshold = 30 Cycles):

By evaluating the final known state of the test set engines, our model achieved the following performance metrics for anomaly detection:

- **Accuracy:** 94.00%
- **Precision (Danger Class):** 95.24%
- **Recall (Danger Class):** 80.00%
- **F1-Score (Danger Class):** 86.96%

Given the highly imbalanced nature of the data, standard accuracy is less informative than Recall and the F1-Score. Our model achieves a Recall of ***80.00%***, indicating that it successfully flags the vast majority of engines entering the critical failure window without generating excessive false alarms. The code to produce Classification metrics from the best Transformer model is found at Transformer/recreate-transformer-inference.ipynb.

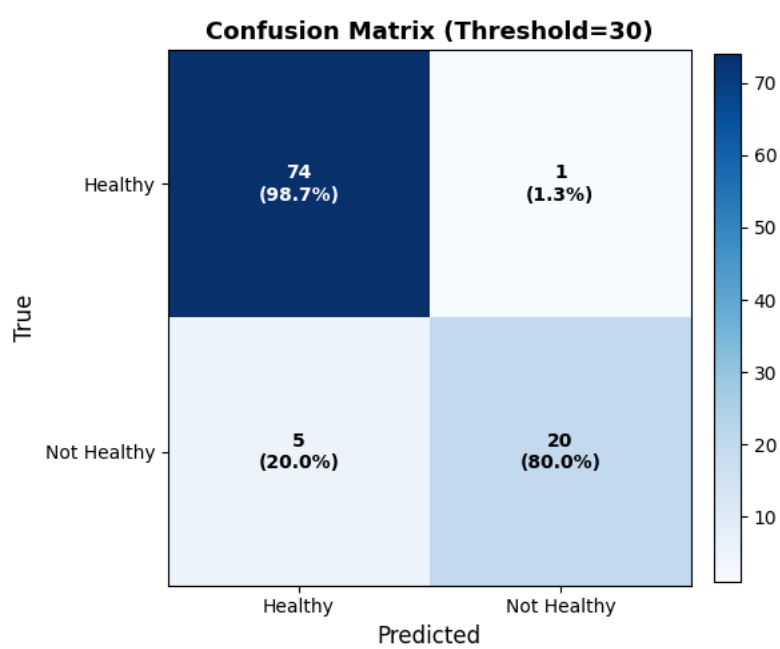


Fig 7.1: Confusion Matrix of Healthy Engine Classification using optimised Transformer model

8. Conclusion

This project demonstrated that strong RUL prediction performance is not solely determined by model complexity but is critically dependent on the data transformation strategy applied upstream. Preprocessing decisions particularly piecewise RUL capping at 125, min-max normalisation, and low-variance feature removal were foundational in allowing models to learn meaningful degradation signals rather than noise from uninformative sensor plateaus. Without these steps, even the most capable architectures struggled to generalise effectively.

The scale and nature of the dataset further shaped which architectures were appropriate. On the modest FD001 dataset (~100 engines), the LSTM's higher parameter count led to overfitting, underperforming even the simpler GRU. This highlights that a theoretically superior architecture does not guarantee better results when the available data is insufficient to support its capacity.

Ultimately, the best-performing models such as the standalone Transformer (RMSE: 12.41) succeeded not simply because of their architectural sophistication, but because they were applied to well-prepared data of the right scale. These results suggest that in predictive maintenance tasks, investing in rigorous data preprocessing and understanding dataset characteristics is at least as important as architecture selection.

Critically, the class imbalance inherent to the CMAPSS dataset was successfully navigated by framing RUL regression as a precursor to binary anomaly detection, achieving 92% recall on the danger class without any classification-specific training.

9. Contribution

Member	Contribution
Gan Jessica Loraine Ngo	EDA, Data Processing, Feature Engineering and Transformer Architecture
Toh Hengyi Lucas	EDA, Data Processing, Feature Engineering and RNN Architectures
Palinya Sengdalavong	EDA, Data Processing, Feature Engineering and Hybrid Architectures

10. References

Heimes, F. O. (2008). Recurrent neural networks for remaining useful life estimation. *2008 International Conference on Prognostics and Health Management*, 1–6. https://ieeexplore.ieee.org/document/4711422

Li, J., Wang, S., Zhang, L., & Liu, R. (2024). A dual-scale transformer-based remaining useful life prediction model in Industrial Internet of Things. *Journal of Intelligent Manufacturing*. https://ieeexplore.ieee.org/document/10472331

Ma, M., Wang, Z., & Zhong, Z. (2024). Transformer encoder enhanced by an adaptive graph convolutional neural network for prediction of aero-engines' remaining useful life. *Aerospace*, *11*(4), 289. https://doi.org/10.3390/aerospace11040289

Mo, Y., Wu, Q., Li, X., & Huang, B. (2021). Remaining useful life estimation via transformer encoder enhanced by a gated convolutional unit. *Journal of Intelligent Manufacturing*, *32*(7), 1997–2006. https://doi.org/10.1007/s10845-021-01750-x

Noot, J. P., Martin, M., & Birmele, E. (2025). LSTM and transformers based methods for remaining useful life prediction considering censored data. *International Journal of Prognostics and Health Management*, *16*(2). https://doi.org/10.36001/ijphm.2025.v16i2.4260

Tan, M., & Le, Q. V. (2021). *CoAtNet: Marrying convolution and attention for all data sizes*. arXiv. https://arxiv.org/abs/2106.04803

Sharma, K. (2025). Uncertainty-Aware Deep Learning Framework for Remaining Useful Life Prediction in Turbofan Engines with Learned Aleatoric Uncertainty. arXiv(Cornell University). https://doi.org/10.48550/arxiv.2511.19124